Лекция

БЧХ и РС коды

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$d \le n-k+1$

1 —

2 Код с пров. на четность

3 код Хэмминга

4 расшир. код Х

$$H \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$(7,4)$

$GF(2)$ - $1+x+x^3$

$\to GF(2^3)$

| | |
|---|---|
| $0$ | $0$ |
| $\alpha^0$ | $1$ |
| $\alpha^1$ | $x$ |
| $\alpha^2$ | $x^2$ |
| $\alpha^3$ | $1+x$ |
| $\alpha^4$ | $x+x^2$ |
| $\alpha^5$ | $1+x+x^2$ |
| $\alpha^6$ | $1+x^2$ |

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 \end{pmatrix}$$

$$\hookrightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$(15, 11)$ $\qquad GF(2) \to GF(2^4)$

$H = (1\ \alpha\ \alpha^2 \ldots \alpha^{14})$ $\qquad p(x) = 1 + x + x^4$

$c(x) + e(x) = b(x)$

$$s = \sum_{i=0}^{n-1} b_i \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i)\alpha^i = \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$$

в поз. $i$ – ошибка $\quad e(x) = x^i, \quad s = \alpha^i$

$e(x) = x^i + x^j, \ i \neq j$

$$\begin{cases} s_1 = \alpha^i + \alpha^j \\ s_2 = f(\alpha^i) + f(\alpha^j) \end{cases} \qquad H = \begin{pmatrix} 1 & \alpha & \ldots & \alpha^{14} \\ \beta_1 & \beta_2 & \ldots & \beta_{15} \end{pmatrix} \qquad \beta_i \in GF(2^m)$$

$f(\alpha) = c\alpha \qquad GF(2)$

$f(\alpha) = \alpha^2 \qquad b(x^2) = b^2(x)$

$f(\alpha) = \alpha^3 \qquad 1^2 = 1 \quad 0^2 = 0$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

$x = \alpha^i \qquad y = \alpha^j \qquad i, j$ – позиции ош.

$$\begin{cases} x + y = s_1 \\ x^3 + y^3 = s_2 \end{cases}$$ при $t_{max} = 2$ ошибка $s_1 \neq 0$

$GF(2)$: $(x+y)^2 = x^2 + y^2 = s_1^2$

$$s_1^2 + xy = \frac{s_2}{s_1}$$

$$\begin{cases} xy = \frac{s_2}{s_1} - s_1^2 \\ x + y = s_1 \end{cases} \xrightarrow{\text{лемма}} z^2 + \sigma_1 z + \sigma_2 = 0$$

$\sigma_1 = s_1 \quad \sigma_2 = \frac{s_2}{s_1} - s_1^2$

$GF(2) \to GF(2^4)$ $\quad p(x) = 1 + x + x^4$

| степень $\alpha$ | полином | двоичн. |
|---|---|---|
| 0 | $1$ | 0001 |
| 1 | $x$ | 0010 |
| 2 | $x^2$ | 0100 |
| 3 | $x^3$ | 1000 |
| 4 | $1+x$ | 0011 |
| 5 | $x+x^2$ | 0110 |
| 6 | $x^2+x^3$ | 1100 |
| 7 | $1+x+x^3$ | 1011 |
| 8 | $1+x^2$ | 0101 |
| 9 | $x+x^3$ | 1010 |
| 10 | $1+x+x^2$ | 0111 |
| 11 | $x+x^2+x^3$ | 1110 |
| 12 | $1+x+x^2+x^3$ | 1111 |
| 13 | $1+x^2+x^3$ | 1101 |
| 14 | $1+x^3$ | 1001 |

$\epsilon$ 1110011000110111

11 0 0011 0111 0111

$S_1 = \alpha^2 + \alpha^9 = \alpha^{11}$

$S_2 = \alpha^6 + \alpha^{22} = \alpha^4$

$$\begin{cases} xy = \dfrac{\alpha^4}{\alpha^{11}} + (\alpha^{11})^2 = \alpha^8 + \alpha^7 = \alpha^{11} \\ x+y = \alpha^{11} \end{cases}$$

$z^2 + \alpha^{11} z + \alpha^{11} = 0$

$f(z)$ $\qquad f(\alpha^2) = f(\alpha^9) = 0$